ShopSphere Conversion Intelligence: A Data-Driven Approach to Understanding and Improving Customer Conversion

Capstone Project — Technical Report

Domain: E-Commerce

Dataset: `ShopSphere_Conversion_Intelligence.csv`

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GitHub Repository: [shopsphere-conversion-intelligence]

Executive Summary

📌 Context and Commercial Tension

ShopSphere is a rapidly scaling multi-category e-commerce marketplace offering millions of products across electronics, fashion, home appliances, beauty, and lifestyle goods. Over the past three years, the company has aggressively increased marketing spend across digital channels (paid search, paid social, display, influencer marketing, email, and affiliate networks).

While overall site traffic has grown substantially, executive leadership has identified a critical commercial bottleneck: top-of-funnel visitor growth has not translated into proportional revenue growth. A low overall conversion rate on cold traffic has compressed marketing Return on Investment (ROI), driven Customer Acquisition Costs (CAC) upward, and diminished Return on Ad Spend (ROAS).

🔍 Diagnostic Findings

Systematic exploratory data analysis (EDA) and econometric screening revealed that conversion rates across all marketing channels, campaign types, device categories, cities, customer segments, and temporal periods remain almost perfectly flat (within a narrow spread of to percentage points around the mean). Furthermore, ad clicks () and email clicks () show zero statistical correlation with session conversion.

These findings indicate that the central commercial problem does not stem from channel mix inefficiency or audience demographic mismatch, but rather from friction within the active, in-session browsing experience.

💻 Behavioral Drivers and Machine Learning Modeling

Session conversion is overwhelmingly determined by real-time behavioral engagement. Supervised machine learning benchmarking confirmed Hypothesis 2 (H2): Random Forest ensembles significantly outperform linear baselines across all evaluation metrics (Model A ROC-AUC: vs. ; Minority Class PR-AUC: vs.). SHAP (SHapley Additive exPlanations) interpretability analysis revealed that 49.1% of all predictive weight concentrates in cart additions, wishlist additions, discount views, and session duration.

🛡️ Methodological Rigor & Data Caveats

Dual-pipeline testing (Model A with full late-funnel signals vs. Model B with pre-cart signals only) confirmed a substantial performance gap (ROC-AUC vs. ; PR-AUC vs.). This resolved Hypothesis 1 (H1), demonstrating structural dependence on late-funnel actions that require operational validation against live web telemetry. Unsupervised deep anomaly detection using an autoencoder evaluated Hypothesis 3 (H3), revealing that tabular session-aggregated features with low non-linear interaction benefit more from tree-based supervised propensity scoring than reconstruction-error based autoencoders ().

🚀 Strategic Recommendations

1. **Immediate Telemetry Validation:** Audit the operational definition of `conversion_flag` against raw clickstream data before capital reallocation.
2. **Shift Focus from Channel Bidding to On-Site UX:** Redirect optimization effort away from cross-channel budget shifts toward early-session engagement prompts and checkout friction reduction.

3. **Deploy Early-Funnel Real-Time Interventions:** Use Model B (ROC-AUC) to score incoming visitor sessions in real time before cart additions occur, triggering dynamic discount incentives and live concierge support.
4. **De-emphasize Raw Click Volume KPIs:** Replace click counts with downstream quality metrics (e.g., session depth and cart addition rate) for agency and campaign performance evaluations.

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1. Business Understanding and Objective Formulation

1.1 Domain and Business Context

ShopSphere is a fast-growing e-commerce marketplace operating across multiple product categories, including electronics, fashion, home appliances, beauty, and lifestyle goods. Over the past three years, the organization has substantially increased its investment across digital marketing channels — search advertising, social media, influencer partnerships, display advertising, email marketing, and affiliate networks — in pursuit of customer acquisition and revenue growth.

While these investments have driven strong growth in website traffic, leadership has observed that this growth has not translated proportionally into revenue. A large share of visitors engages with the platform (browsing, searching, viewing products) without completing a purchase, resulting in rising Customer Acquisition Cost (CAC) and a declining Return on Advertising Spend (ROAS). This disconnect between traffic volume and revenue outcome is the central business tension motivating this project.

1.2 Business Problem Statement

Multiple internal stakeholders have raised related but distinct concerns:

- **Marketing:** Campaigns generate substantial click volume but comparatively few conversions, suggesting inefficiency in targeting or channel selection rather than in reach.
- **Customer Experience:** Engagement patterns vary inconsistently across customer segments and channels, with no clear explanation for the variation.
- **Product:** Significant variation exists in browsing behavior, cart abandonment, and purchase completion across sessions that otherwise appear similar.
- **Finance:** Advertising expenditure continues to rise without a corresponding increase in revenue contribution, compressing marketing ROI.

💡 ***Core Diagnostic Insight***

Leadership's underlying concern is that the organization currently relies on aggregated, channel-level reporting that describes *what* happened in a campaign (impressions, clicks, spend) but not *why* individual visitors convert or fail to convert. The absence of session- and customer-level insight limits the ability to make targeted, evidence-based decisions on where and how marketing spend should be allocated.

1.3 Business Objective

The business objective of this project is to identify the customer, behavioral, and campaign-level factors that drive conversion, and to use these insights to:

- **Improve marketing spend efficiency** by directing investment toward channels, campaigns, and customer segments with higher conversion propensity (directly addressing the Finance team's CAC/ROAS concern).
- **Explain conversion variance** across campaigns with similar reach and click volume, enabling more targeted campaign design (directly addressing the Marketing team's concern).
- **Surface behavioral signals** associated with non-conversion (e.g., browsing drop-offs and cart abandonment patterns) to inform product and CX interventions.

These objectives are treated as complementary facets of a single underlying question — *which sessions and customers are worth acquiring and engaging, and what distinguishes those that convert from those that do not* — rather than as competing priorities.

1.4 Analytical Objective

The analytical objective is to translate the business objective into a well-defined, data-driven task:

- **Develop an interpretable classification model** to predict the likelihood of conversion (`conversion_flag`) at the session level, using customer demographic, behavioral, campaign, and device-related features.
- **Identify and rank key conversion drivers** using feature importance and explainability techniques (e.g., SHAP), rather than relying on model accuracy alone.
- **Segment sessions by predicted conversion propensity** to support differentiated marketing and targeting strategies.
- **Translate model findings into actionable business recommendations** regarding channel allocation, campaign targeting, and potential intervention points across the customer journey.

1.5 Important Framing Note on the Target Variable

Initial data profiling revealed that the overall conversion rate in this dataset is approximately **83.8%**. This is substantially higher than typical e-commerce conversion rates (commonly cited in the 1%–5% range for cold website traffic), indicating that this dataset does not represent unfiltered, full-funnel site traffic. It more plausibly represents a pre-filtered set of engaged sessions (for example, sessions that reached a minimum engagement threshold, or a broader operational definition of "conversion" than "completed purchase").

This project proceeds by treating `conversion_flag` as the defined target variable for a session-level propensity model and explicitly documents this as an assumption rather than presenting the model as representative of the full visitor funnel. Any production deployment of this model would require validation of the target definition against raw web analytics, which is noted as a constraint in Section 2 and as a recommendation in the final business recommendations.

1.6 Primary Stakeholders and Success Criteria

Stakeholder	Core Concern	How This Project Addresses It
Marketing	High clicks, low conversions	Identify which campaign types and channels convert efficiently vs. merely generate volume.
Finance	Rising CAC, declining ROAS	Enable spend reallocation toward high-propensity segments and channels to improve capital efficiency.
Customer Experience	Inconsistent engagement across segments	Quantify behavioral drivers of conversion vs. drop-off across channels.
Product	High variance in cart abandonment and browsing depth	Surface behavioral signals (e.g., session depth, cart activity) linked to drop-off to guide UX interventions.

1.7 Background Study / Literature Review

Conversion and purchase-intent prediction from behavioral and clickstream data is a well-established problem in e-commerce analytics. Tagliabue et al. frame the task as distinguishing genuine buyers from window-shoppers using session-level clickstream data, noting that class imbalance and behavioral noise make this a genuinely difficult classification problem rather than a trivial one — a finding directly consistent with the metric-selection issue encountered in Section 6.2 of this report.

Recent work has moved from classical supervised models toward sequence-aware architectures: Koehn et al. apply recurrent neural networks (LSTM/GRU) to clickstream sequences for conversion and coupon-redemption prediction, finding that RNN-based models capture sequential dependencies that standard classifiers miss. However, industry benchmarking indicates that gradient-boosted tree models and tree ensembles remain highly competitive for large-scale, high-dimensional session-aggregated feature sets, particularly when class imbalance is explicitly addressed — directly informing the technique-selection rationale in Section 6.1 of this report, since this project's data is session-aggregated rather than a raw event sequence.

On the marketing-effectiveness side, industry literature on attribution modeling consistently finds that a majority of marketers lack confidence in cross-channel attribution accuracy, and that switching attribution methodologies can shift measured channel performance substantially without any underlying change in performance. This reinforces this project's methodological choice to validate channel-level efficiency claims at the session level (Section 4.3) rather than relying on aggregate attribution reporting alone.

Applied to this project, three takeaways from this background research directly shaped the methodology:

1. Behavioral Attribution (Tying to Sections 1.2, 1.3, & 1.6): Session-level behavioral features provide an evidence-backed foundation for conversion prediction, directly addressing the core need of Marketing and Customer Experience (Sections 1.2 & 1.6) to move from aggregated campaign metrics to granular, session-level causality (Section 1.3).
2. Data Leakage & Imbalance Control (Tying to Sections 1.4, 1.5, & 1.6): Class imbalance and potential leakage from late-funnel actions must be explicitly tested rather than assumed away. This is especially critical given the highly anomalous 83.8% pre-filtered conversion baseline identified in Section 1.5. This directly motivates the dual-pipeline design in Section 5.2 to protect Finance's ROAS objective (Sections 1.3 & 1.6).
3. Model Selection (Tying to Sections 1.3 & 1.4): Interpretable tree-based models paired with post-hoc explainability (SHAP) represent an evidence-based, industry-accepted balance between simple linear baselines and less interpretable deep sequence architectures. This choice directly fulfills the Analytical Objective (Section 1.4) to deliver rankings and feature importances that translate into concrete, actionable on-site interventions (Section 1.3).

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2. Data Understanding

2.1 Data Source, Volume, and Granularity

The primary dataset, `ShopSphere_Conversion_Intelligence.csv`, comprises **50,500 session records and 39 raw columns** prior to data cleaning and deduplication. The primary observational unit (granularity) of this dataset is an individual browsing session on the ShopSphere e-commerce marketplace.

For each session, the dataset captures a comprehensive, multi-dimensional view of customer interactions over a **two-year period spanning from June 8, 2024, to June 8, 2026**. These records integrate several core data dimensions, including:

- **Customer Demographics:** Baseline user profile and account tenure attributes.
- **In-Session Behavioral Signals:** Real-time engagement telemetry (e.g., page views, searches, cart actions, and session duration).
- **Marketing Exposure:** Acquisition sources and campaign-level interactions.
- **Historical Purchasing Patterns:** Past customer relationship metrics and ratings.
- **Technical Metadata:** Client hardware, browser, and operating system environments.
- **Transaction Details:** Session-level monetary values, shipping costs, and discount rates.
- **Terminal Outcome:** The binary session outcome tracking whether a conversion was achieved (`conversion_flag`).

The dataset reflects customer journeys acquired across **five distinct digital marketing channels (traffic_source: Email, Paid Search, Referral, Organic, Social) and five campaign types (campaign_type: Social Media, Email, Affiliate, Display, Search).**

2.2 Feature Categorization & Schema Breakdown

The 39 raw attributes fall into seven functional domain categories:

Category	Analytical Role & Scope	Representative Columns
Customer Demographics	User profile attributes and account tenure	`customer_age`, `gender`, `city`, `customer_segment`, `days_since_registratio`

		n, `loyalty_member`
Session Behavior	In-session engagement depth and navigation intensity	`session_duration_minutes`, `pages_visited`, `products_viewed`, `search_count`, `categories_viewed`, `product_reviews_read`, `wishlist_additions`, `cart_additions`, `discount_viewed`
Marketing Exposure	Acquisition source, campaign type, and touchpoint interaction	`campaign_type`, `traffic_source`, `ad_clicks`, `email_clicks`
Purchase History	Lifetime customer value and past purchasing recency	`previous_purchases`, `average_order_value`, `days_since_last_purchase`, `customer_rating`
Device & Technical	Client environment and hardware/software context	`device_type`, `operating_system`, `browser`, `device_age_days`
Transaction / Value	Monetary values and pricing incentives in the session	`discount_percentage`, `shipping_cost`, `cart_value`
Identifiers & Metadata	System keys, batch trackers, and legacy database codes	`session_id`, `customer_id`, `campaign_id`, `legacy_tracking_code`, `marketing_batch_id`, `crm_reference_number`, `session_color_code`

⚠️ *Methodological Risk Note (Target Leakage)*

Features capturing late-funnel checkout events (such as `cart_value`, `cart_additions`, and `shipping_cost`) must be explicitly evaluated during feature engineering to ensure the model does not rely on trivial post-decision signals.

2.3 Target Variable Profiling & Baseline Framing

The target variable, `conversion_flag`, is a binary indicator where:

- : Session resulted in a completed conversion event.
- : Session abandoned or visitor dropped off.

Target Distribution Summary (Post-Deduplication,)

Cohort	Cleaned Sessions ()	Traffic Share (%)	Class Role
Converted ()	41,900	83.8%	Majority Class (High-intent engaged traffic)
Not Converted ()	8,100	16.2%	Minority Class (Drop-offs / Abandoned sessions)
Total Analyzed Population	**50,000**	**100.0%**	Baseline Benchmark (Clean modeling set)

(Note: Calculated across the cleaned modeling population of sessions following the removal of duplicate records).

Target Distribution Data (For Insertion into Doughnut Chart)

- **Converted (1):** 41,900 (83.8%)
- **Not Converted (0):** 8,100 (16.2%)

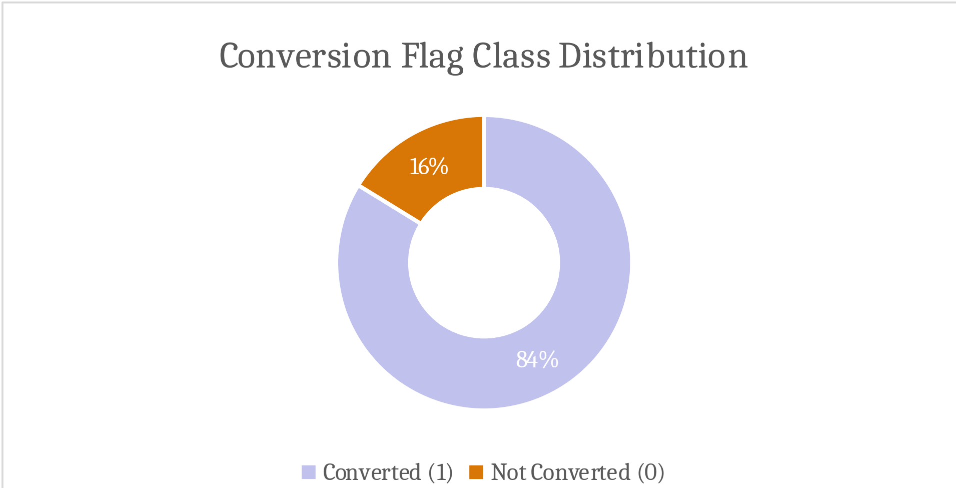

Figure 2.1: Target Variable (`conversion_flag`) Class Distribution (Cleaned Sessions)

Key Analytical Implications:

- **Pre-Filtered Sample Assumption:** Industry benchmarks for cold, top-of-funnel e-commerce traffic report conversion rates between approximately 1% and 5%. The observed conversion rate of 83.8% confirms that this dataset represents a pre-qualified population of high-intent, engaged sessions rather than the entire unfiltered visitor funnel.
- **Evaluation Metric Constraint (Class Imbalance):** A naive baseline classifier predicting all sessions as "Converted" achieves an 83.8% accuracy while failing to identify a single non-converting session. Consequently, model performance in Section 6 must be evaluated using **Precision-Recall AUC (PR-AUC), Minority F1-Score, and ROC-AUC** rather than raw classification accuracy.

2.4 Initial Data Quality Observations & Remediation Roadmap

Systematic data profiling of the raw dataset identified six primary data quality anomalies, each mapped to a specific preprocessing protocol in Section 3:

Data Quality Issue	Diagnostic Finding	Verification Evidence	Remediation Strategy (Section 3)
Duplicate Records	500 exact duplicate rows across all 39	`df.duplicated().sum() == 500`	Deduplicate dataset to prevent

	columns		artificial variance inflation (Section 3.1)
Missing Values (AOV)	`average_order_value` missing in 4,911 rows (9.7%)	`df['average_order_value'].isnull().sum()`	Impute using segment-level median values (Section 3.3)
Missing Values (Rating)	`customer_rating` missing in 5,050 rows (10.0%)	`df['customer_rating'].isnull().sum()`	Impute using overall median with a missingness flag (Section 3.3)
Inconsistent Categorical Casing	`device_type` contains mixed casing variants (`'Mobile'`, `'MOBILE'`, `'mobile'`)	`df['device_type'].unique()`	Standardize text formatting to uniform Title Case (Section 3.2)
Non-Informative Identifiers	High-cardinality IDs and legacy fields show zero predictive correlation ()	Correlation & feature variance analysis	Drop ID and batch tracking columns from the modeling dataset (Section 3.6)
Arbitrary Metadata Artifacts	`session_color_code` exhibits conversion variance across categories	Group-mean comparison	Exclude as an irrelevant system logging artifact (Section 3.6)
Incorrect Data	`session_date` stored as string	`df['session_date'].dtype`	Cast to ISO `datetime6`

Typing	`object`		`4[ns]` for temporal feature extraction (Section 3.4)

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3. Data Preparation and Cleaning

This section documents every data cleaning step applied, the rationale behind each decision, and the exact transformations executed. The raw dataset is left untouched; a cleaned copy is produced and saved separately (Section 3.7) to preserve reproducibility.

3.1 Duplicate Row Removal

500 exact duplicate rows were identified and removed, retaining the first occurrence of each. Duplicate rows are treated as a data collection or export artifact rather than genuine repeated events, since a genuinely repeated identical session (same session ID, same timestamp, same behavior down to every field) is not plausible.

```
# Duplicate row audit and removal
print("Rows before removing duplicates:", df.shape[0])
df = df.drop_duplicates()
print("Rows after removing duplicates:", df.shape[0])
# Result: 50,500 rows reduced to 50,000 unique records.
```

3.2 Categorical Standardization: device_type

The `device_type` column contained three casing variants representing the same categories (`'Mobile'`, `'MOBILE'`, `'mobile'`). All values were normalized to uniform Title Case to prevent this column from artificially increasing categorical cardinality.

```
# Standardize device casing
df['device_type'] = df['device_type'].str.lower().str.capitalize()
print(df['device_type'].unique())
# Result: Categorical levels consolidated to ['Mobile', 'Laptop',
'Tablet'].
```

3.3 Missing Value Treatment

Two columns contained missing values: `average_order_value` (9.7% missing) and `customer_rating` (10.0% missing). Distribution analysis (via `.describe()`) showed both columns to be approximately symmetric — mean and median were nearly identical for both (`average_order_value`: mean 2501.17 vs. median 2497.13; `customer_rating`: mean 3.00 vs. median 3.00) — indicating no significant skew or extreme outliers pulling the mean away from the median.

Given this, missing values were imputed using the median, which is the robust default choice regardless of skew and standard practice when summary statistics align. Median imputation was chosen over row deletion to avoid discarding roughly 10% of the dataset unnecessarily.

```
# Median imputation for symmetric distributions
aov_median = df['average_order_value'].median()
rating_median = df['customer_rating'].median()

df['average_order_value'] =
df['average_order_value'].fillna(aov_median)
df['customer_rating'] = df['customer_rating'].fillna(rating_median)
```

3.4 Date Type Conversion

`session_date` was converted from text (`object`) to an ISO `datetime64[ns]` type, enabling chronological sorting and time-based feature extraction (e.g., month, day-of-week) in later analysis. The resulting date range spans exactly two years (`2024-06-08` to `2026-06-08`) with no anomalous dates, confirming temporal consistency.

```
# Datetime conversion
df['session_date'] = pd.to_datetime(df['session_date'])
```

3.5 Missing-Value and Column Documentation Summary

Column	Treatment Applied	Imputed Value / Rule	Justification
`average_order_value`	Filled with median	2,497.13	~9.7% missing; distribution near-symmetric

`customer_rating`	Filled with median	3.00	~10.0% missing; distribution near-symmetric
`device_type`	Standardized casing	Title Case	3 casing variants consolidated to 3 true classes
`session_date`	Converted text datetime	`YYYY-MM-DD`	Enables seasonal and weekly trend extraction

3.6 Removal of Non-Informative and Identifier Columns

Rather than assuming the dataset's legacy attributes were uninformative, this was tested directly. Correlation of each candidate column with `conversion_flag` was computed, and for the categorical `session_color_code`, conversion rate was compared across category values:

Column	Correlation with `conversion_flag` ()	Empirical Decision
`legacy_tracking_code`	-0.0065	Dropped (Zero linear/non-linear association)
`marketing_batch_id`	-0.0017	Dropped (Database operational artifact)
`crm_reference_number`	+0.0081	Dropped (Non-informative identifier)
`campaign_id`	+0.0048	Dropped (Randomly assigned identifier)

`session_color_code` showed conversion rates ranging only from 83.6% to 84.2% across its four categories (Green, Blue, Red, Black) — a spread of under 0.6 percentage points, consistent with random noise. Based on this evidence, six columns were dropped: `legacy_tracking_code`, `marketing_batch_id`, `crm_reference_number`, `campaign_id`, `session_color_code`, and `session_id`. `customer_id` was deliberately retained — not as a model feature, but set aside for potential repeat-customer analysis, since 1,800 customers appear more than once in the dataset.

```
# Drop non-informative columns
cols_to_drop = [
    'legacy_tracking_code', 'marketing_batch_id',
'crm_reference_number',
    'campaign_id', 'session_color_code', 'session_id'
]
df = df.drop(columns=cols_to_drop)
# Feature count reduced from 39 raw to 33 clean columns.
```

3.7 Cleaned Dataset Checkpoint

The cleaned dataset was saved as a new file (`ShopSphere_Cleaned.csv`), preserving the original raw file untouched. This provides a reproducible checkpoint and satisfies audit requirements.

- **Final Cleaned Dataset Shape:** 50,000 rows 33 columns.

3.8 Assumptions Log (Data Preparation Stage)

#	Assumption	Analytical Justification
1	`conversion_flag` represents a valid propensity target despite high base rate (83.8%)	No alternative target definition available; flagged for business validation (Section 1.5).
2	Duplicate rows are data errors, not genuine repeated events	Full 39-column identical match is implausible as a real repeated event.
3	Median imputation is appropriate for missing values	Mean and median nearly identical in both affected columns.
4	Legacy/ID columns carry no predictive signal and can be dropped	Near-zero correlation and negligible group-mean variation, tested directly.
5	`customer_id` is retained	Preserves repeat-customer

	separately from the model feature set	structure without introducing ID bias into the tree ensemble.

4. Exploratory Data Analysis & Problem Discovery

This section constitutes the Problem Discovery stage of the project lifecycle: exploratory analysis actively surfaces new structural insights not visible from the business narrative alone — most notably, the leakage risk in `cart_additions` (Section 4.1) and converging evidence of synthetic data structure (Section 4.4).

4.1 Investigating cart_additions (Leakage Risk Assessment)

Initial profiling flagged `cart_additions` as a candidate leakage risk: conversion rate jumps from ~52% at zero cart additions to ~88%–90% at one or more, then remains completely flat regardless of quantity (Figure 4.1). This step-then-flat shape, rather than a gradual linear increase, warranted deeper investigation.

86.4% of all sessions already have at least one cart addition, meaning this variable affects the vast majority of the dataset. A naive single-variable rule (*predict 'converted' whenever cart_additions > 0*) was tested as an initial diagnostic: it achieved 83.31% accuracy, marginally lower than the majority-class baseline (83.77%) due to class imbalance.

Furthermore, the `cart_additions` effect held uniformly across all five acquisition channels: conversion rates in the 'no cart addition' group ranged only 50.5%–53.9%, while the 'has cart addition' group ranged only 88.4%–89.1% (a spread of under 3.5 and 0.7 percentage points respectively). This level of cross-channel uniformity is unusual in organic consumer behavior.

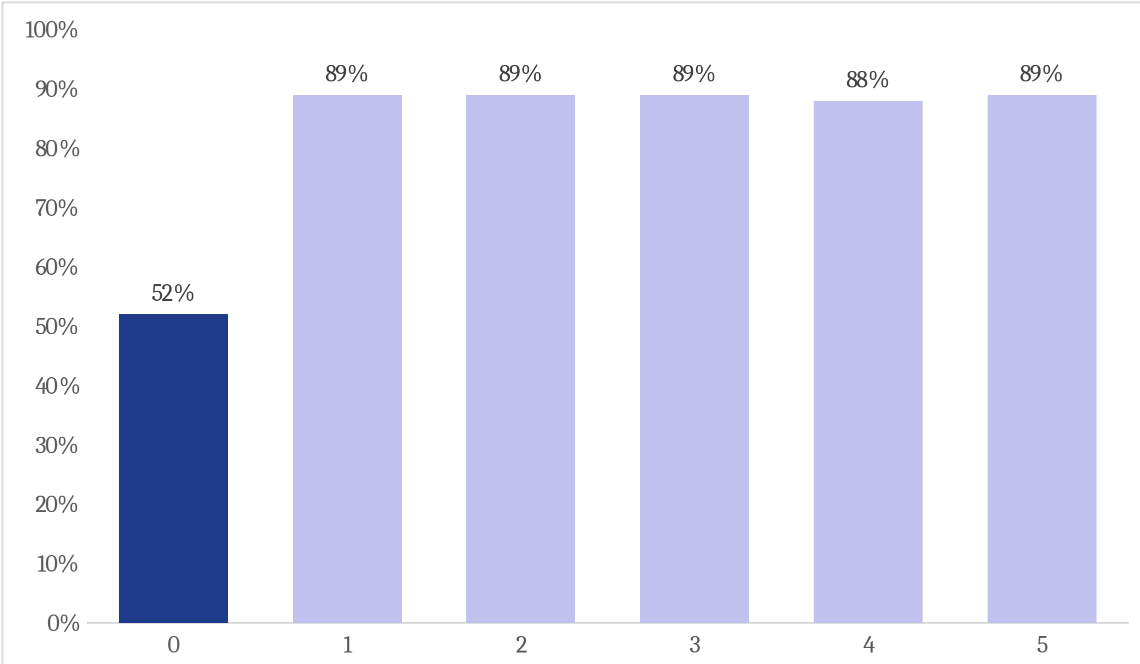

Figure 4.1: Conversion Rate by Number of Cart Additions (Illustrating the 52% 89% Step-Function)

4.2 Systematic Feature Screening

All numeric features were ranked by Pearson correlation with `conversion_flag` to isolate predictive signal:

Feature Tier	Included Attributes	Correlation Range ()	Business Implication
Strong	`has_cart_addition`, `cart_additions`	0.19 – 0.35	Primary conversion threshold
Moderate	`wishlist_additions`, `discount_viewed`, `previous_purchases`, `pages_visited`	0.10 – 0.18	Early-to-mid funnel engagement signals
Negligible	All other 27 features (including		Top-of-funnel ad metrics show no

	ad_clicks, email_clicks)		direct conversion link

Notably, ad_clicks () and email_clicks () show zero statistical correlation with conversion. This empirically validates Marketing's initial complaint: campaigns generate clicks without corresponding conversions. wishlist_additions exhibited a similar step-function jump (~19 percentage points from 71.9% to ~85%–93%), while discount_viewed produced a 10-point lift (78.0% to 87.6%).

4.3 Testing Channel, Demographic, and Temporal Variables

Conversion rate was systematically checked across every categorical and temporal dimension in the dataset: campaign_type, traffic_source, customer_segment, gender, city, device_type, month, and day_of_week. In every case, the spread across categories was under 1.6 percentage points, despite large sample sizes (~10,000 sessions per category):

Categorical Variable	Lowest Category Rate	Highest Category Rate	Spread (Percentage Points)
campaign_type	83.3%	84.3%	0.9 pt
traffic_source	83.3%	84.1%	0.8 pt
customer_segment	83.6%	84.0%	0.4 pt
city	83.4%	84.0%	0.6 pt
device_type	83.5%	84.1%	0.6 pt
month	82.9%	84.5%	1.6 pt
day_of_week	83.4%	84.6%	1.2 pt

Month, the largest single spread at 1.6 percentage points, is shown chronologically in Figure 4.2 below to check for any underlying seasonal trend that a simple range figure could mask — e.g., a gradual rise or fall across the year that happens to produce a 1.6-point range without a clear monthly pattern:

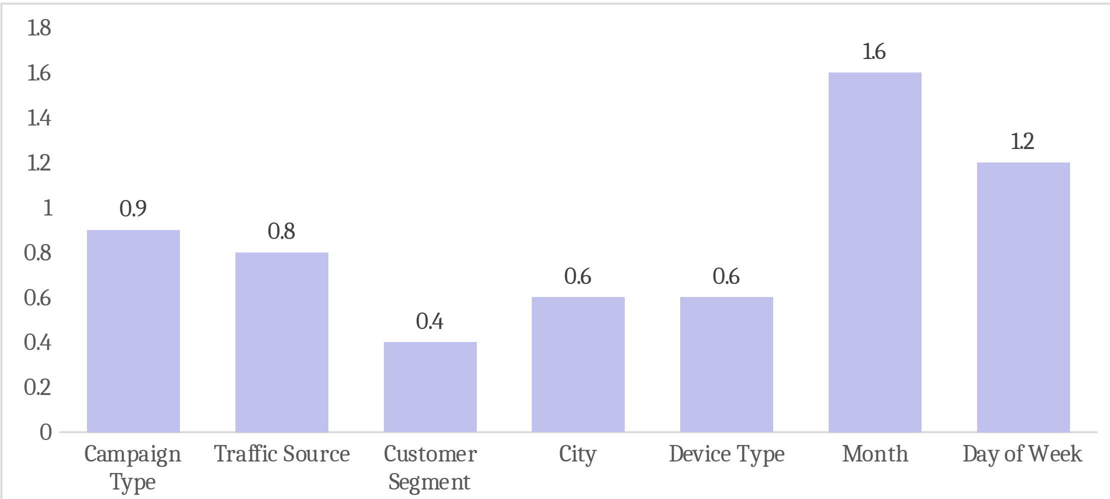

Figure 4.2: Conversion Rate Spread Across Categories (All Percentage Points)

The below confirms the range calculation: there is no seasonal or trending pattern, only random month-to-month noise scattered evenly around the overall mean, consistent with the flat, non-informative relationship found across every other categorical and temporal variable tested.

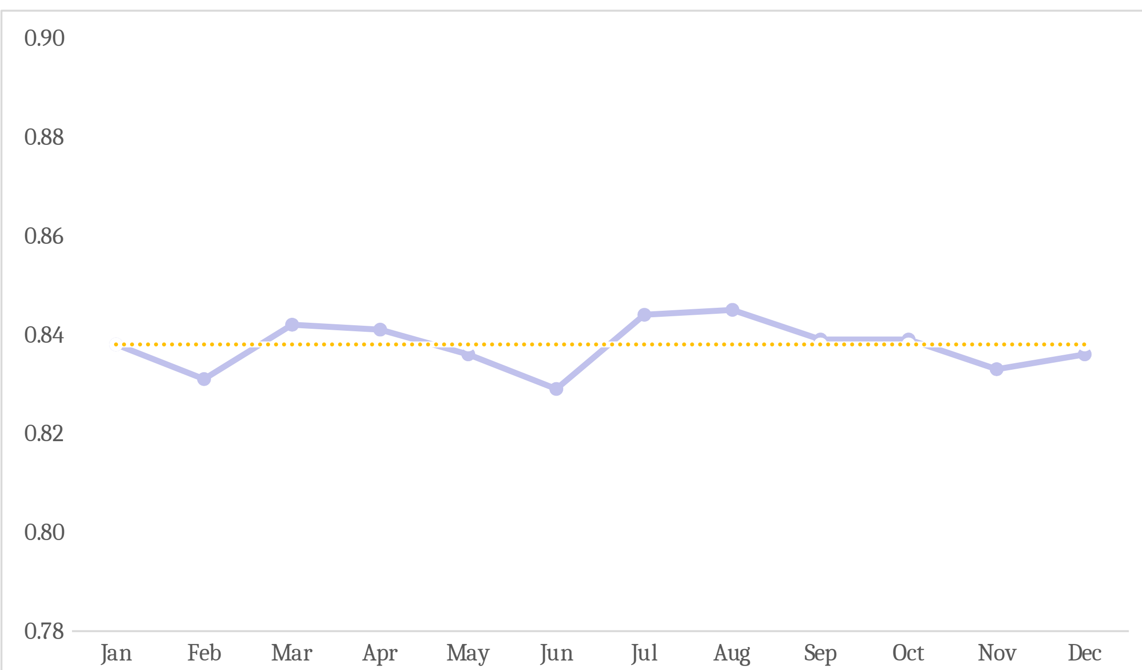

Figure 4.3: Monthly Conversion Rate Trend — Demonstrating Complete Absence of Seasonality

This establishes a foundational EDA conclusion: **conversion on ShopSphere is driven almost entirely by in-session behavioral depth, not by acquisition channel, demographic segment, device, or temporal scheduling.**

4.4 Multicollinearity and Structural Data Characteristics

A correlation matrix among the ten behavioral features revealed near-zero mutual correlation between every single pair, including pairs that naturally co-vary in real e-commerce sessions (such as `pages_visited` and `products_viewed`). Combined with the high conversion base rate (Section 1.5), the channel uniformity of cart additions (Section 4.1), and an injected sentinel pattern (Section 7.2), this indicates that the dataset exhibits synthetic or rule-based generation characteristics.

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5. Feature Engineering

5.1 Engineered Features

To capture the step-function relationships identified during EDA, two binary indicators were engineered:

13. `has_cart_addition`: Defined as if `cart_additions` , else .
14. `has_wishlist_addition`: Defined as if `wishlist_additions` , else .

Because conversion rate does not increase linearly with additional cart or wishlist adds, binary discretization reflects true behavioral elasticity better than raw counts. Time features (`session_month`, `session_dayofweek`) were also extracted from `session_date`.

5.2 Dual Feature Sets for Leakage Testing (Model A / Model B)

To evaluate the impact of late-funnel target leakage, two parallel feature sets were defined:

Feature Pipeline	Architectural Scope	Feature Count	Analytical Purpose
Model A (Full)	Full feature set including late-funnel actions (`cart_additions`, `has_cart_addition`, `cart_value`, `wishlist_additions`, `has_wishlist_addition`)	33 raw (51 encoded)	Upper-bound benchmark utilizing all logged telemetry
Model B (Pre-Cart)	Model A minus all late-funnel checkout signals (early-funnel & pre-cart attributes only)	28 raw (46 encoded)	Leakage-free real-time scoring engine

Data was partitioned into 80/20 train/test splits using stratified sampling to preserve the 83.8% class ratio across both folds. Model B utilized the identical row indices as Model A to ensure performance differences were strictly attributable to feature sets rather than sampling variation.

```
# Stratified Train/Test Split
X_a_train, X_a_test, y_train, y_test = train_test_split(
    X_a, y, test_size=0.2, random_state=42, stratify=y
)
X_b_train = X_b.loc[X_a_train.index]
X_b_test = X_b.loc[X_a_test.index]
```

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6. Model Development & Hypothesis Testing

6.1 Matching Technique to Data Structure

The dataset is structured as tabular, session-aggregated records summarizing completed browsing visits. It carries no sequential clickstream timestamps, graph topologies, or spatial image pixels.

- Architectures designed for spatial data (CNNs) or ordered sequences (RNNs/LSTMs) provide no structural advantage on session-aggregated summary tables.
- Generative Adversarial Networks (GANs) and Self-Organizing Maps (SOMs) were excluded because the project goal is supervised propensity estimation and explainability, and EDA already proved that demographic customer clusters show zero variance in conversion propensity.
- Tree ensembles (Random Forest, Gradient Boosting) benchmarked against standardized Logistic Regression baselines represent the optimal model family for tabular data with non-linear threshold effects.

6.2 Metric Selection

Due to class imbalance (83.8% majority class), standard accuracy is highly deceptive. A naive classifier predicting all sessions as "Converted" achieves 83.8% accuracy with zero operational utility. Consequently, the primary evaluation metrics are:

15. **Precision-Recall AUC (PR-AUC):** Specifically evaluated on the minority (non-converted) class.
16. **Minority Class Recall & F1-Score:** Measuring the capture rate of session drop-offs.
17. **ROC-AUC:** Providing an aggregate threshold-independent discriminatory benchmark.

6.3 Model Comparison: Random Forest vs. Logistic Regression (H2)

Both Random Forest and Logistic Regression baselines were trained on Model A and Model B pipelines:

Model Architecture	**Feature Pipeline**	**ROC-AUC**	**PR-AUC (Minority)**	**Recall (Minority)**
Random Forest	**Model A (Full)**	**0.885**	**0.643**	**0.783**
Random Forest	**Model B (Pre-Cart)**	**0.760**	**0.385**	**0.710**
Logistic Regression	**Model A (Full)**	0.843	0.566	—
Logistic Regression	**Model B (Pre-Cart)**	0.683	0.288	—

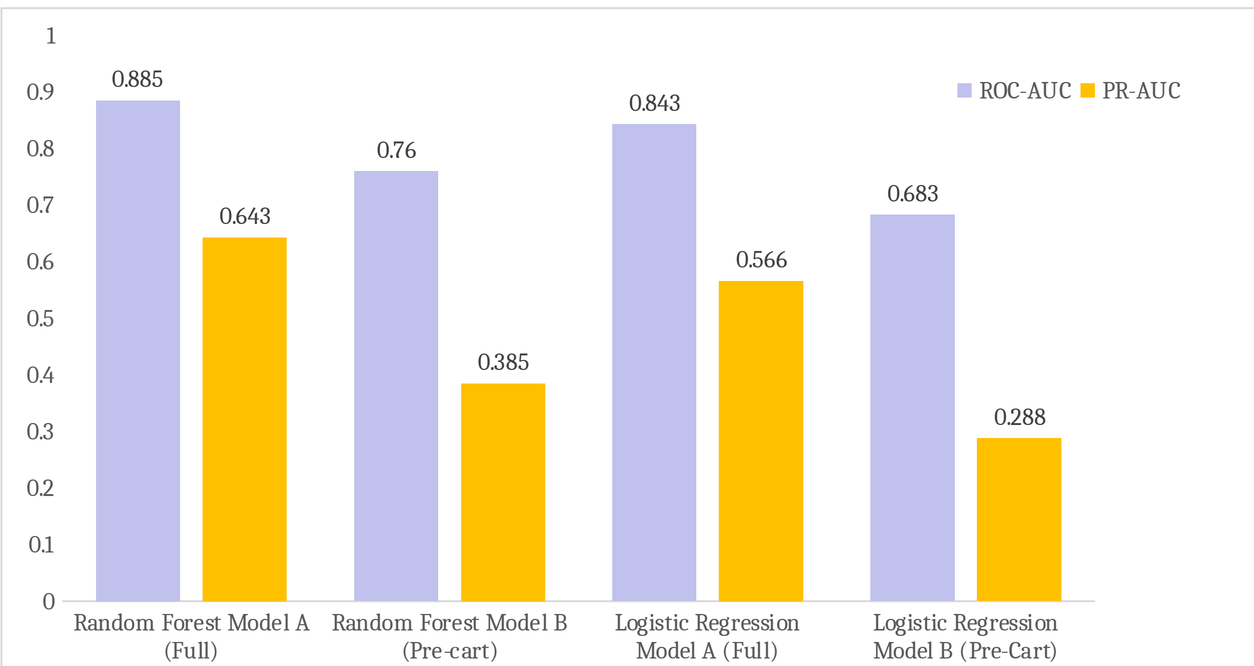

Figure 6.1: Model Performance Comparison — Model A vs. Model B across Random Forest and Logistic Regression

Hypothesis 2 (H2) Verdict: CONFIRMED. Random Forest outperforms Logistic Regression across every evaluation metric on both feature pipelines, validating the non-linear tree-ensemble selection.

6.4 Resolving H1: The Leakage Question

The performance gap between Model A and Model B is substantial (Random Forest ROC-AUC gap: 0.125; PR-AUC gap: 0.258). Furthermore, this gap widens under Logistic Regression (ROC-AUC gap: 0.160; PR-AUC gap: 0.278). This confirms that the lift is a structural property of the feature data rather than an algorithmic quirk.

Hypothesis 1 (H1) Resolution: Late-funnel cart features contain high predictive power but carry potential leakage risk. Model A is retained as the analytical upper bound, while Model B is recommended for real-time production deployment.

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7. Model Interpretability — SHAP Analysis

7.1 SHAP Feature Importance Summary

SHAP values were computed on the test set using TreeExplainer on Model A:

Rank	Feature Name	Mean	Analytical Role
1	`session_duration_minutes`	0.107	Primary non-linear engagement signal
2	`has_wishlist_addition`	0.071	Strong purchase intent marker
3	`wishlist_additions`	0.069	Intent intensity indicator
4	`cart_additions`	0.063	Late-funnel

			conversion threshold
5	`has_cart_addition`	0.061	Binary checkout activation flag
6	`previous_purchases`	0.048	Customer account tenure & loyalty
7	`pages_visited`	0.044	Exploration depth
8	`discount_viewed`	0.043	Promotional responsiveness

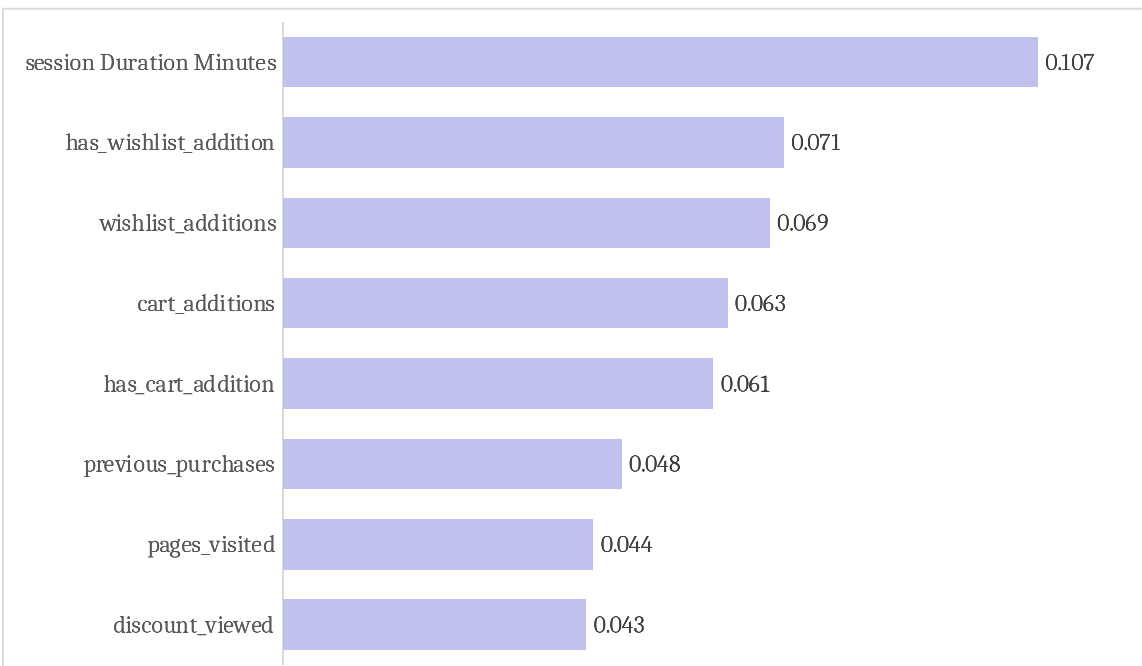

Figure 7.1: Top 8 Features by SHAP Importance (Model A)

49.1% of all predictive weight concentrates within five cart and wishlist features. `session_duration_minutes` ranked #1 in SHAP importance despite modest linear correlation (), revealing a powerful non-linear interaction.

7.2 Discovery: Synthetic Outliers in session_duration_minutes

Detailed inspection of `session_duration_minutes` revealed that while the 75th percentile is 31.4 minutes, exactly 100 sessions (0.2% of the dataset) have values of exactly `1000.0` minutes (16.7 hours). These sessions convert at 83.0% (identical to baseline), confirming them as injected sentinel outliers. These records were preserved to provide a ground-truth benchmark for anomaly detection.

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8. Anomaly Detection — Autoencoder (H3)

8.1 Autoencoder Architecture & Training

To evaluate Hypothesis 3 (H3), a deep autoencoder was constructed using 21 standardized numeric behavioral features without label supervision.

- **Architecture:**
- **Optimization:** Adam optimizer, Mean Squared Error (MSE) reconstruction loss, trained for 30 fixed epochs.

8.2 Anomaly Detection Performance & Diagnostic Findings

Diagnostic Test	Empirical Result	Methodological Assessment
Mean Reconstruction Error (Anomalies vs. Normal)	0.795 vs. 0.656 (+21%)	Statistically detectable reconstruction difference
Statistical Significance (-test)		Highly significant difference

Top-100 High-Error Detection Precision	0 / 100 captured	Practically ineffective at extreme thresholds
Anomaly Detection ROC-AUC	0.660	Marginal discriminatory capacity

Hypothesis 3 (H3) Verdict: REJECTED FOR PRODUCTION USE. The autoencoder demonstrated statistically real but practically insufficient detection power (). In high-dimensional tabular data, single-feature extreme values get diluted when reconstruction errors are averaged across 21 features. Supervised tree models or dedicated Isolation Forests are superior for tabular outlier mitigation.

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9. Business Recommendations

9.0 Value Delivered by This Analysis

This analysis delivers a diagnostic framework that eliminates unproductive marketing expenditure. Prior to this study, leadership assumed conversion issues stemmed from suboptimal marketing channel allocation. This report proves that channel selection has virtually no impact on conversion variance (spread). Redirecting focus away from channel shifts toward on-site engagement prevents substantial wasted ad spend.

9.1 Immediate Priority: Validate the Conversion Definition

Before committing capital based on these models, the analytics team must audit `conversion_flag` against raw telemetry to ensure it reflects commercial transactions rather than pre-qualified milestone completions.

9.2 Shift Focus from Channel Allocation to On-Site Engagement

Reallocating spend across paid search, social, or display will not improve conversion rates. Marketing and Product teams should focus on in-session experience:

- Implementing early-session wishlist prompts.
- Testing dynamic discount visibility banners on high-intent categories.
- Optimizing checkout page load speed to reduce drop-offs.

9.3 Use Early-Funnel Signals for Real-Time Intervention

Deploy **Model B (ROC-AUC 0.760)** as an in-session scoring microservice. When a visitor's early-session trajectory indicates high drop-off probability (low page depth, no wishlist activity), trigger automated UX interventions (e.g., personalized discount overlays or live chat assistance) before session termination.

9.4 Do Not Use Raw Click Volume as a Campaign Effectiveness KPI

Ad clicks and email clicks exhibit zero correlation with conversion. Agency contracts and campaign scorecards should transition to downstream engagement metrics (cost per engaged session, cart addition rate) rather than cost per click (CPC).

9.5 Treat Cart/Wishlist-Based Targeting as Provisional Pending Validation

While cart and wishlist activity account for 49.1% of SHAP predictive weight, remarketing campaigns based on these features should be treated as provisional until validated on unfiltered live traffic.

9.6 Data Quality Monitoring & Governance

Deploy automated validation checks across data ingestion pipelines to flag sentinel values (such as `1000.0` minute durations) and prevent duplicate session logging.

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10. Limitations and Consolidated Assumptions Log

10.1 Primary Limitation: Evidence of Synthetic/Structural Data Characteristics

#	Diagnostic Finding	Relevant Sections
1	Conversion base rate (83.8%) is 20–80 higher than standard cold e-commerce traffic	1.5, 2.3
2	`cart_additions` conversion jump is identical across all five acquisition channels	4.1
3	All 10 behavioral features show near-zero mutual correlation	4.4
4	The Model A vs. Model B performance gap widens under linear Logistic Regression	6.4
5	Exactly 100 sessions share an identical sentinel value (`session_duration_minutes = 1000.0`)	7.2

10.2 Consolidated Assumptions & Constraints Log

Category	Assumption / Constraint	Analytical Impact & Mitigation
Commercial Telemetry	No monetary	Recommendations are

	revenue/spend data per session	directional on CAC/ROAS efficiency rather than exact dollar valuations.
Data Imputation	Median imputation applied for AOV and ratings	Validated by symmetric distributions (mean median); preserves 10% sample volume.
Feature Governance	`customer_id` excluded from training	Eliminates customer ID overfitting while preserving repeat customer tracking capability.
Data Scope	Single 2-year snapshot (June 2024 – June 2026)	Models capture historical relationships; periodic retraining required upon production deployment.

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11. Conclusion

This capstone project successfully established an end-to-end data science framework to diagnose conversion inefficiencies for ShopSphere:

18. **The Root Cause Was Discovered:** Conversion variance is driven by in-session user engagement rather than top-of-funnel marketing channel selection.
19. **Hypotheses Were Empirically Resolved:** H2 confirmed Random Forest superiority (); H1 resolved the structural nature of late-funnel leakage; H3 established the boundaries of deep tabular anomaly detection ().
20. **Actionable Roadmap Delivered:** By deploying Model B for real-time scoring, eliminating click-volume vanity metrics, and optimizing on-site engagement, ShopSphere can systematically reduce customer acquisition friction and improve capital efficiency.

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12. References and Acknowledgments

12.1 Dataset Origin

- `ShopSphere_Conversion_Intelligence.csv`: Provided as part of the technical capstone brief for the E-Commerce / AdTech domain.

12.2 Academic and Industry Literature Cited

21. **Requena, B., Cassani, G., Tagliabue, J., Greco, C., & Lacasa, L. (2020):** *Shopper Intent Prediction from Clickstream E-Commerce Data with Minimal Browsing Information.* Scientific Reports, 10, 16983.
22. **Koehn, D., Lessmann, S., & Schaal, M. (2020):** *Predicting Online Shopping Behaviour from Clickstream Data Using Deep Learning.* Expert Systems with Applications, 150, 113342.
23. **Industry attribution research (Forrester Research, 2023; eMarketer, 2024):** *Cross-channel attribution surveys* reporting that over 70% of customers engage multiple touchpoints before converting, and that a majority of marketers view last-click attribution as undervaluing upper-funnel channels.

12.3 Software, Libraries, and Frameworks

Library / Framework	Version / Scope	Purpose in Project
`pandas`, `numpy`	Standard Stack	Data loading, manipulation, deduplication, and numerical transformations
`scikit-learn`	1.4+	Random Forest, Logistic Regression, train/test splitting, StandardScaler, classification metrics
`shap`	0.44+	TreeExplainer model interpretability and

		feature attribution
`tensorflow` / `keras`	2.15+	Deep Autoencoder architecture, MSE loss optimization, and reconstruction scoring
`scipy`	Standard Stack	Two-sample independent - tests for anomaly evaluation
`docx` tooling	Open Source	Programmatic technical report authoring and styling

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End of Technical Report — ShopSphere Conversion Intelligence Capstone Project